

# DynaSpace Smartphone PPS Looming Stimulus

## Spectral and Spatial Feature Audit

PPS-kit derived report

2026-06-28

### Executive Decision

The smartphone DynaSpace stimulus should not be copied as an opaque WAV into PPS-kit. It should be decomposed into a parameterized dry burst train plus a 3DTI/SOFA trajectory renderer. The useful smartphone-study features are: a broadband Gaussian white-noise burst train, a strong but not extreme rising distance/intensity envelope, the 6.4 m to 0.2 m constant-bearing trajectory, log-spaced tactile anchors, and room/externalization cues visible in the low interaural coherence of the baked WAV. Those are worth adopting as explicit parameters.

The features that PPS-kit/3DTI should handle instead of imitating by raw spectral hacks are HRTF spectral filtering, ILD, ITD, near-field corrections, future trajectory angles, and any controlled reverberation/direct-to-reverberant behavior. The current PPS-kit proxy file does not yet match the raw DynaSpace audio: it is continuous rather than burst-train-like, much higher in RMS level, less bright, highly interaurally coherent, and has a stable opposite-signed ILD/ITD convention relative to the DynaSpace raw files. The target 3DTI framework is a better conceptual approach, but the present proxy render is not yet a faithful DynaSpace-style implementation.

### Scope and Inputs

This audit isolates the raw spectral, temporal, level, and binaural features in the DynaSpace Android audio files used by Roussel et al.'s smartphone PPS validation study [1]. The original raw WAV files were measured from a private local archival clone of the public DynaSpace source repository [2]. The raw files are not redistributed here. PPS-kit artifacts in this folder are derived metrics, plots, TeX, and PDF only.

The compared audio files are:

- DynaSpace raw looming: `bursttrainlooming.wav`, SHA256 prefix `610c9a9c2232`. Full hash is stored in `dynaspace_audio_feature_metrics.json`.
- DynaSpace raw fixed: `bursttrainfixe.wav`, SHA256 prefix `56f8afff1b09`. Full hash is stored in `dynaspace_audio_feature_metrics.json`.
- PPS-kit proxy looming: `assets/preloads/rousseau_2025_dynaspace_mobile_pps/02_looming_stimuli/looming_DynaSpace_proxy_looming_burst_train.wav`.
- PPS-kit proxy fixed: `assets/preloads/rousseau_2025_dynaspace_mobile_pps/02_looming_stimuli/looming_DynaSpace_proxy_fixed_640_cm_burst_train.wav`.

### Search and Consensus Method

Search date: 2026-06-28. I used web searches across the smartphone PPS paper, DynaSpace app, Hobeika burst-train lineage, Canzoneri PPS looming task, auditory looming intensity literature, auditory localization/distance reviews, and 3DTI documentation. The requested Consensus search was attempted through tool discovery, but no Consensus connector was exposed in this Codex session; the available discovered tools were unrelated Gmail tools. Therefore, this report uses a documented web/primary-source evidence base instead of a Consensus API result.

Table 1: Search iterations and evidence role.

| Feature class                          | Search iteration                                                                                         | Evidence used in this report                                                                                                                                                                                                                                                                 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Smartphone PPS parameters              | Queries for Roussel, DynaSpace, 640 cm, 20 cm, 210 cm/s, LISTEN, and tactile anchors.                    | Roussel et al. report a DynaSpace Android app, a 3250 ms auditory stimulus, a fixed 640 cm source, a looming source from 640 cm to 20 cm at ear level and 210 cm/s, azimuth 45 degrees in left hemispace, and tactile delays T1–T6 at 105, 1625, 2385, 2765, 2955, and 3050 ms [1].          |
| Burst-train source                     | Queries for Hobeika, Gaussian white noise bursts, 30 ms, 10 ms rise/fall, 65 ms intervals.               | The lineage stimulus is a Gaussian white-noise burst train with short bursts and ramps; 30 ms bursts plus 65 ms interval predicts a 95 ms onset spacing, matching the 33 measured DynaSpace peaks [3].                                                                                       |
| Basic PPS effect                       | Queries for dynamic sounds, boundaries of peripersonal space, approaching auditory stimuli.              | Canzoneri et al. show that dynamic sounds can reveal spatially dependent audio-tactile facilitation and PPS boundaries, with approaching sounds producing the stronger tactile RT pattern [4].                                                                                               |
| Rising level and looming warning value | Queries for auditory looming bias, rising sound intensity, warning cue, amygdala, phasic alertness.      | Rising intensity is supported as a warning/alerting cue and is central to auditory looming salience [5, 6].                                                                                                                                                                                  |
| Localization cue inventory             | Queries for auditory localization ITD, ILD, HRTF, bandwidth, reverberation, direct-to-reverberant ratio. | Reviews identify ITD, ILD, HRTF spectral filtering, bandwidth, intensity, reverberation, and direct-to-reverberant ratio as spatial/distance cues [7, 8].                                                                                                                                    |
| 3DTI capability                        | Queries for 3D Tune-In Toolkit, SOFA, HRTF, distance attenuation, near-field, BRIR, reverberation.       | 3DTI supports SOFA/HRIR convolution, ITD customization, distance attenuation, near-field corrections, movement smoothing, and BRIR/Ambisonic reverberation in the full toolkit [9].                                                                                                          |
| Local PPS-kit implementation           | Repository search for FABIAN, SOFA, gain law, nearest HRIR, reverb, and generated proxy assets.          | PPS-kit stores FABIAN as the standard redistributable HRTF resource, records LISTEN as excluded pending license audit, and the current Python reference renderer uses nearest HRIR lookup, 1/r gain when enabled, peak normalization unless loudness policy is enabled, and reverb disabled. |

## Measured Raw Signal Inventory

Measurements were generated by `measure_dynaspace_audio.py`. Burst detection used a 20 ms RMS envelope, 2.5 ms hop, 3 dB peak prominence, and at least 55 ms spacing. Binaural lag was measured by cross-correlation within  $\pm 1$  ms; the sign is a script convention and should not be treated as a biological left/right truth until a channel-sign probe is run.

Table 2: Compact derived feature summary.

| Metric                         | Dyna looming | Dyna fixed  | PPS looming | PPS fixed   |
|--------------------------------|--------------|-------------|-------------|-------------|
| Sample rate (Hz)               | 48000        | 48000       | 44100       | 44100       |
| Duration (s)                   | 3.850        | 3.850       | 3.850       | 3.850       |
| Peak (dBFS)                    | -3.19        | -33.48      | -0.91       | -0.91       |
| RMS (dBFS)                     | -37.28       | -53.25      | -22.95      | -16.66      |
| Crest factor (dB)              | 34.10        | 19.77       | 22.03       | 15.74       |
| Detected burst peaks           | 33           | 33          | 0           | 0           |
| Median burst IOI (ms)          | 95.0         | 95.0        | n/a         | n/a         |
| Active p90-p10 level span (dB) | 18.20        | 8.09        | 29.22       | 0.88        |
| Active level slope (dB/s)      | 4.31         | -1.16       | 9.40        | 0.02        |
| Spectral centroid (Hz)         | 10948        | 9319        | 8094        | 8115        |
| 95% rolloff (Hz)               | 19910        | 19846       | 15558       | 15558       |
| ILD L-R (dB)                   | -6.34        | -4.26       | 12.72       | 12.68       |
| Cross-corr lag (us)            | 187.5        | 187.5       | -362.8      | -362.8      |
| IACC abs                       | 0.229        | 0.132       | 0.695       | 0.696       |
| Moving 100 ms IACC range       | 0.043–0.269  | 0.043–0.228 | 0.668–0.718 | 0.666–0.719 |

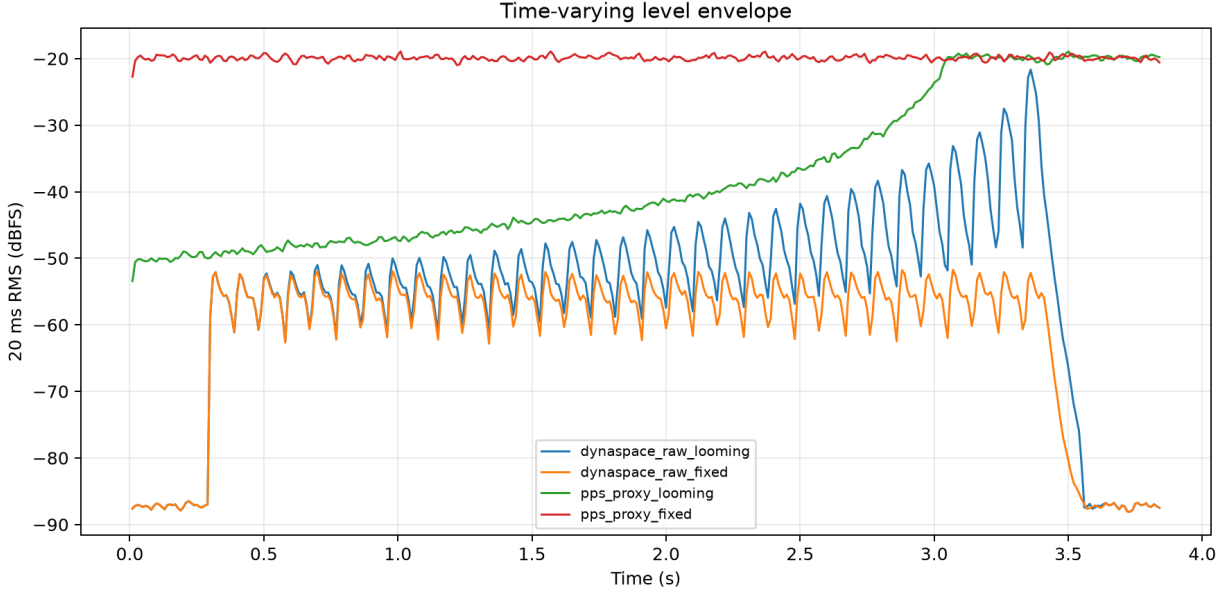


Figure 1: 20 ms RMS envelope comparison. DynaSpace raw looming is pulsed and rises over the trial; PPS-kit proxy looming is continuous and uses a stronger smooth gain ramp.

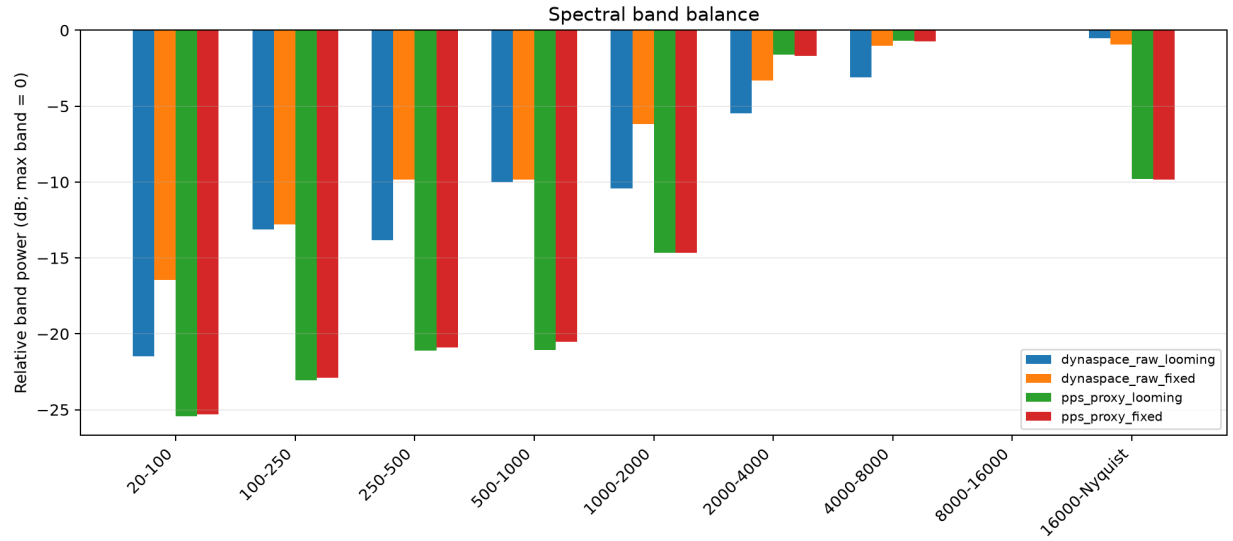


Figure 2: Spectral band comparison. The DynaSpace raw looming file is brighter, with more relative high-frequency energy and a higher centroid than the current PPS-kit proxy.

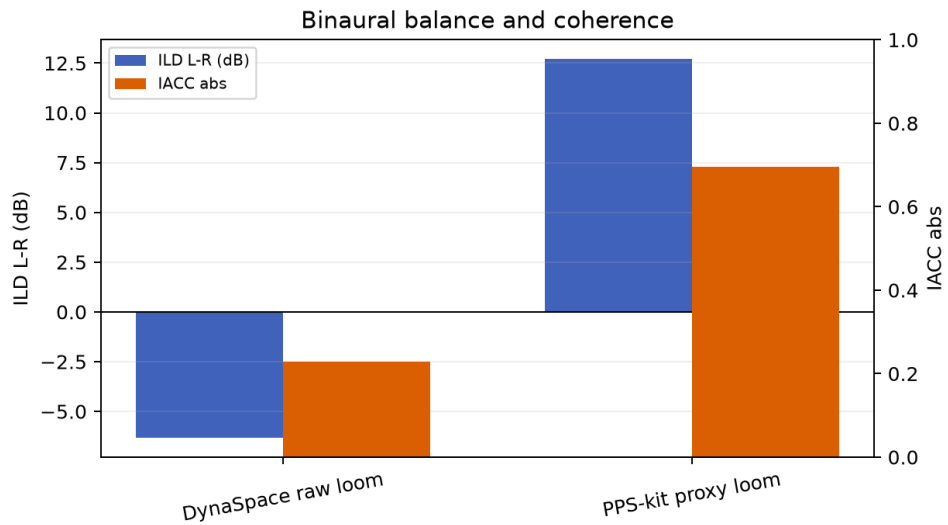


Figure 3: Binaural summary. The raw DynaSpace looming file has lower interaural coherence and opposite-signed/much smaller ILD than the current PPS-kit proxy.

# One-to-One Feature Audit

Table 3: Feature-by-feature decision matrix.

| Feature                             | Raw DynaSpace evidence                                                                                                                                                                                           | Spatialization or PPS rationale                                                                                                                                                      | PPS-kit / 3DTI status                                                                                                                                                                                             | Decision                             |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| Broadband Gaussian/noise source     | Raw files are broadband stereo noise bursts. DynaSpace looming centroid is 10.95 kHz and 95% rolloff is 19.91 kHz.                                                                                               | Broadband content makes HRTF spectral filtering, ILD, and localization cues more available than narrowband tones [7]. It also matches the Hobeika PPS lineage [3].                   | PPS proxy uses generated white noise, so the class is correct, but its spectrum is less bright and continuous. 3DTI can spatialize any dry source; the dry source spectrum should be controlled before rendering. | <b>Adopt with controls</b>           |
| Burst train temporal structure      | Raw looming and fixed files both contain 33 detected peaks, median IOI 95 ms. Measured half-prominence widths are 55 ms looming and 70 ms fixed, inflated by ramps/reverb/detector relative to dry burst design. | The burst train creates repeated onsets while preserving a continuous trial. Onsets are strong for alerting and localization, and the 95 ms spacing agrees with the Hobeika lineage. | Current PPS proxy has no comparable burst structure under peak-prominence detection. This is the clearest missing feature.                                                                                        | <b>Adopt</b>                         |
| Burst ramps and click control       | The lineage uses short burst ramps; measured raw files have high crest factors but no clipping.                                                                                                                  | Ramps avoid click artifacts while retaining salient attacks.                                                                                                                         | PPS generator should create dry bursts with explicit raised-cosine rise/fall parameters, not rely on random noise fluctuations.                                                                                   | <b>Adopt</b>                         |
| Overall duration and silence window | Raw files are 3.85 s. Paper reports a 3250 ms auditory stimulus preceded by 300 ms silence and followed by a randomized 1000–1400 ms silence at trial level [1].                                                 | Timing window matters because tactile anchors sample distance over the approach.                                                                                                     | PPS profile matches 3.85 s proxy duration and anchor logic. Keep meta-data explicit because paper-level and raw-file-level timings are not identical descriptions.                                                | <b>Adopt with controls</b>           |
| Distance trajectory                 | Paper trajectory: 640 cm to 20 cm at ear level, constant speed 210 cm/s, left hemisphere azimuth 45 degrees; anchors map to 640, 320, 160, 80, 40, and 20 cm [1].                                                | This is the main PPS geometry. It separates temporal expectancy from spatial proximity through fixed-vs-looming contrast.                                                            | PPS template already encodes 6.4 m to 0.2 m, 2.105 m/s, signed -45 degrees, and anchors. 3DTI is better than a baked file for future angles.                                                                      | <b>Already better in 3DTI target</b> |

| Feature                          | Raw DynaSpace evidence                                                                                                                        | Spatialization or PPS rationale                                                                                                                   | PPS-kit / 3DTI status                                                                                                                                                                                                                                             | Decision                             |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| Rising intensity / distance gain | DynaSpace raw looming active p90-p10 span is 18.20 dB; PPS proxy is 29.22 dB. Raw fixed is much quieter overall than raw looming.             | Rising level is a core auditory looming warning cue [5, 6]; distance intensity is also a primary distance cue [8, 7].                             | PPS proxy has a stronger smooth gain ramp and is much higher RMS. Current Python path applies 1/r amplitude gain when gain law is enabled, then peak-normalizes unless loudness policy disables it. Native 3DTI can express direct-path attenuation more cleanly. | <b>Adopt with controls</b>           |
| Fixed comparator level           | Raw fixed peak is -33.48 dBFS and RMS is -53.25 dBFS, far lower than raw looming.                                                             | A far fixed comparator may intentionally represent source distance, but it also risks making the main effect partly a loudness/salience contrast. | PPS fixed proxy is loud and continuous, not DynaSpace-like. For exact replication, preserve far-level comparator; for clean inference, add a matched-level control profile.                                                                                       | <b>Do not adopt blindly</b>          |
| High-frequency brightness        | DynaSpace looming centroid is 10.95 kHz; PPS proxy is 8.09 kHz. Raw 95% rolloff is near 19.9 kHz, proxy near 15.6 kHz.                        | High frequencies strengthen ILD and pinna/HRTF spectral cues; broad bandwidth improves localization [7].                                          | 3DTI should supply HRTF coloration, but the dry source should remain bright enough. Add optional spectrum matching/shelf only as a dry-source setting, not as a fake spatializer.                                                                                 | <b>Adopt with controls</b>           |
| HRTF spectral filtering          | Paper reports Max/MSP Spat and non-individual LISTEN HRTFs for direct sound and early reflections [1].                                        | HRTFs carry head/pinna/torso spectral cues for externalized spatial location [7].                                                                 | PPS-kit uses FABIAN SOFA as a redistributable standard; LISTEN is excluded pending license audit. Full 3DTI/SOFA rendering is a stronger, parameterized approach than baking the raw spectral coloration.                                                         | <b>Already better in 3DTI target</b> |
| ILD                              | DynaSpace looming global ILD L-R is -6.34 dB; moving-window ILD ranges -7.78 to 0.48 dB. PPS proxy global ILD is +12.72 dB and nearly static. | ILD is a main lateralization cue, especially at high frequencies and near-field positions [7].                                                    | 3DTI should implement ILD through HRTF, distance, head shadow, and near-field corrections. Current proxy sign/magnitude mismatch means coordinate/channel convention must be audited before using the profile for left/right claims.                              | <b>Adopt with controls</b>           |

| Feature                           | Raw DynaSpace evidence                                                                                                                                                     | Spatialization or PPS rationale                                                                                                                                                      | PPS-kit / 3DTI status                                                                                                                                                                   | Decision                             |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| ITD / cross-ear lag               | DynaSpace raw lag is +187.5 us; PPS proxy lag is -362.8 us by the same script convention.                                                                                  | ITD is a principal horizontal cue, especially for lower-frequency components; broadband bursts provide onset information too [7].                                                    | Full 3DTI supports customized ITD and propagation. The Python proxy uses nearest HRIR frames and does not prove correct signed ITD. Add a channel-sign validation file.                 | <b>Adopt with controls</b>           |
| Interaural coherence / IACC       | DynaSpace looming IACC abs is 0.229 globally and 0.043–0.269 in 100 ms windows. PPS proxy is much more coherent at about 0.695 globally.                                   | Low interaural coherence is consistent with reflected/reverberant bin-audal energy and can aid externalization, but excessive decorrelation can weaken precise lateral localization. | 3DTI with BRIR/Ambisonic reverb is a better controlled way to add this than arbitrary decorrelation. Current PPS profile has reverb disabled.                                           | <b>Adopt with controls</b>           |
| Early reflections and late reverb | Paper reports direct sound, early reflections, and late reverberation in the smartphone stimuli [1]. Raw low IACC supports the presence of non-anechoic spatial rendering. | Reverberation contributes distance/externalization information; direct-to-reverberant ratio is a major distance cue [8, 7].                                                          | Full 3DTI supports BRIR/Ambisonic reverberation [9]; current PPS Python reference config declares reverb disabled. Add an optional DynaSpace-room profile rather than always-on reverb. | <b>Adopt</b>                         |
| Direct-to-reverberant ratio       | Not directly measured from the stereo WAV without the dry/render components, but low IACC and paper methods imply a baked direct-plus-reflection mixture.                  | DRR can provide distance information that is less dependent on source level than raw intensity [8].                                                                                  | 3DTI can model this better if a room/BRIR configuration is explicit. Current proxy has no explicit DRR target.                                                                          | <b>Adopt with controls</b>           |
| Near-field distance cues          | Endpoint is 20 cm, inside near field.                                                                                                                                      | Near-field ILD/head-shadow changes become important close to the head; source distance changes are perceptually larger near the body.                                                | 3DTI full high-quality path includes near-field corrections [9]; current Python proxy is a simplified reference render. Use native 3DTI for publication-grade near-field stimuli.       | <b>Already better in 3DTI target</b> |
| Trajectory flexibility            | Raw WAV is a single baked 45-degree left-hemisphere stimulus.                                                                                                              | Future studies need different approach angles and maybe curved trajectories.                                                                                                         | PPS-kit/3DTI is clearly better here because the trajectory is metadata-driven. Do not encode spatial path as immutable audio.                                                           | <b>Already better in 3DTI target</b> |
| Sample rate / bit depth           | DynaSpace raw files are 48 kHz, 16-bit PCM. PPS proxy files are 44.1 kHz. Paper text reports 44.1 kHz source preparation, while Android raw files are 48 kHz.              | Sample rate affects top-band content and exact asset playback, but it is not itself a spatial cue.                                                                                   | Match the HRTF/SOFA/render chain sample rate and document resampling. Use 48 kHz only for exact Android playback replication, not because it is inherently better.                      | <b>Do not adopt blindly</b>          |

| Feature                  | Raw DynaSpace evidence                                                                                                                  | Spatialization or PPS rationale                                                                                                     | PPS-kit / 3DTI status                                                                                                                                                                  | Decision                    |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| Peak/RMS/crest quality   | Raw looming peak - 3.19 dBFS, no clipping, RMS -37.28 dBFS, crest 34.10 dB. PPS proxy peak -0.91 dBFS, RMS -22.95 dBFS, crest 22.03 dB. | High crest is expected for sparse burst trains. Hidden peak normalization can destroy intended absolute or relative level policies. | PPS should disable hidden peak normalization in calibrated mode and store SPL/RMS targets. Raw DynaSpace quality is usable, but its fixed-vs-looming level asymmetry must be declared. | <b>Adopt with controls</b>  |
| Phone/transducer chain   | Paper used an Android phone and headphones; the raw file alone does not encode the full playback chain.                                 | Smartphone output and headphone response can affect level and spectrum, but this is apparatus, not a spatial rendering principle.   | PPS-kit should not mimic unknown phone coloration unless validating Android playback specifically. Use measured audio-interface/headphone calibration instead.                         | <b>Do not adopt blindly</b> |
| Tactile vibration timing | Paper vibration: 155 Hz, about 50 ms; source profile records Android code timing details.                                               | This affects PPS RT alignment, not audio spatial quality.                                                                           | Keep DynaSpace timing anchors and vibration metadata in the study profile, but do not mix haptic waveform design into audio spectral matching.                                         | <b>Adopt with controls</b>  |

## Signal Quality Assessment

The DynaSpace raw looming WAV is technically clean for a behavioral stimulus: it has no clipping, large headroom, high crest factor consistent with a pulse train, and a stable 33-burst temporal pattern. Its level is low on average because most of the file is sparse/noisy and because distance/reverberation were baked into the render. The raw fixed file is also clean, but its peak and RMS are much lower than the looming file. That fixed-vs-looming level gap is not a file-quality problem; it is a design property that can become an interpretive problem if a future study tries to attribute all facilitation to spatial proximity rather than level/salience.

The PPS-kit proxy files are also technically clean, but they are not DynaSpace-equivalent. They are continuous broadband noises, not burst trains; they are peak-normalized close to full scale; the looming proxy has a stronger gain ramp than the raw DynaSpace looming file; and the proxy binaural image is much more coherent. This makes the present proxy useful as a runnable placeholder, but not a final smartphone-style replication stimulus.

## Best-of-Both-Worlds Generation Protocol

1. Generate a dry, mono Gaussian white-noise burst train: 33 bursts, 30 ms dry burst duration, 10 ms raised-cosine rise/fall, 65 ms silent interval, median peak spacing 95 ms, seed recorded in metadata. Keep a source-spectrum option that can target the raw DynaSpace brightness without pretending that spectrum matching is spatial rendering.
2. Use the DynaSpace trajectory profile already encoded in PPS-kit: 6.4 m to 0.2 m, 2.105 m/s, ear level, constant-bearing left approach, tactile anchors at 105, 1625, 2385, 2765, 2955, and 3050 ms. Preserve the ability to change azimuth/elevation/endpoints for future studies.
3. Render with native 3DTI high-quality mode when available. Use SOFA HRIRs, HRIR interpolation, customized ITD/head model, propagation delay, 6 dB-per-distance-doubling direct-path attenuation, and near-field corrections. Use FABIAN as the redistributable default; use LISTEN only in private or after a license audit.
4. Add an optional DynaSpace-room/externalization layer: early reflections plus late reverberation or BRIR/Ambisonic reverb. Tune it with QC targets rather than by ear: lower IACC than the dry proxy, documented direct-to-reverberant settings, and retained direct-source lateralization.



5. Provide two comparator modes. A *DynaSpace replication* mode keeps the far fixed source at 640 cm with the original-style distance/level logic. A *clean causal* mode adds matched-level controls so spatial proximity, temporal expectancy, and loudness/salience can be separated.
6. Add render QC acceptance checks for every generated WAV: no clipping; declared peak/RMS/SPL policy; 33 detected bursts and 95 ms median IOI; active looming level span near the chosen target; channel-sign validation; expected ILD/ITD direction for the requested azimuth; IACC/DRR report if reverb is enabled; and all trajectory anchors stored in sidecar JSON.

## Implementation Consequences for PPS-kit

- The next toolkit change should be a `dynaspace_burst_train` source generator, not another pre-rendered WAV copy.
- The DynaSpace profile should expose two level-policy presets: `smartphone_replication` and `matched_level_control`.
- The renderer should prefer native 3DTI for DynaSpace-style publication stimuli; the Python SOFA reference path is acceptable for preview/proxy but should be flagged as simplified.
- Reverb/BRIR should be optional but first-class. The original smartphone stimulus used reflections/reverb, and the measured low IACC shows that current PPS proxy externalization is not equivalent.
- Before changing left/right signs, run a channel-orientation audit. The raw files and current proxy disagree in ILD/ITD sign, and that can be a coordinate convention issue rather than a true stimulus mismatch.

## Bottom Line

It is worth incorporating several DynaSpace stimulus features into PPS-kit, but only as explicit, testable generation parameters. The strongest additions are the burst-train source, DynaSpace-level trajectory anchors, controlled rising level, and optional reflection/reverb/externalization layer. The features that 3DTI should own are spatial cues: HRTF spectrum, ILD, ITD, near-field behavior, propagation delay, distance attenuation, and arbitrary future trajectories. The raw smartphone WAV is best treated as an audit target and validation reference, not as the renderer.

## References

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